# Learn to Hack Hackathon

# Group 1 – Matrix Not Found

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https://roh00t.github.io/MatrixNotFound/

# Problem Statement

Buildings are a major contributor to carbon emissions. We built an **ML model** that **predicts a building's energy use intensity** (site EUI) from known factors, such as:

- Floor area and year built
- Energy Star rating
- Facility type and location
- Climate: temperatures, degree days, precipitation and snow

Buildings that use far more energy than the model expects for similar buildings are the ones leaders should **retrofit first**.

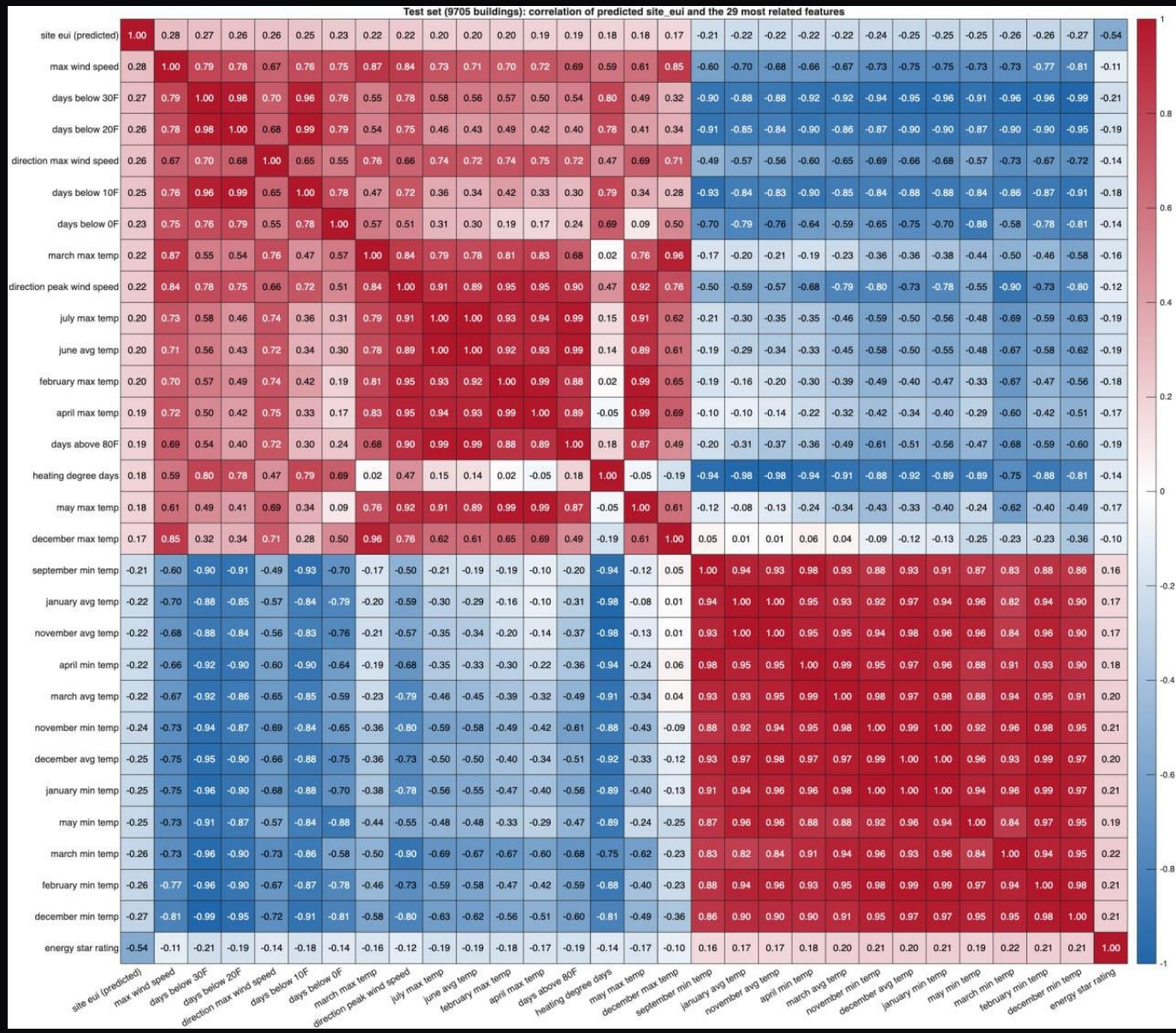


# Exploratory Data Analysis

**75,757** training building-years; **9,705** test buildings.

- The test set is one **unseen year** (year 7) with no State 6 buildings, so we validate by holding out a whole year.
- Energy Star rating is 35% missing: we filled gaps with **training-only medians** and added "was missing" flags (no rows dropped).

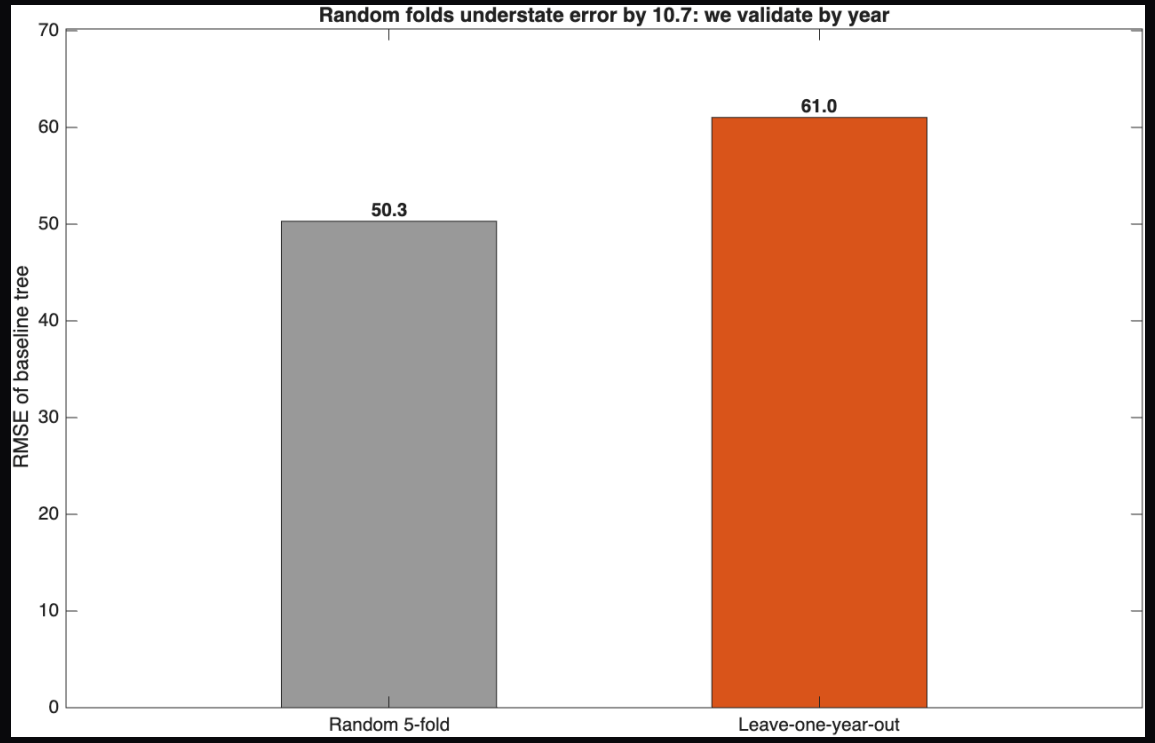
On the test set, **Energy Star rating** has the strongest correlation with site EUI (**−0.54**). Climate variables form tightly correlated blocks.



# Model Architecture

**Gradient-boosted regression trees** (MATLAB **LSBoost**, the native XGBoost-family method): mixed numeric and categorical data, no scaling.

- **Tree** baseline → **LSBoost v1** (all features) → **v2** (shrinkage) → **v3** (feature selection: drop 36 monthly temperatures)
- Engineered features: building age, floor area × heating/cooling degree days, Energy Star × floor area
- **Leave-one-year-out** validation, with cleaning refit inside every fold (no leakage)



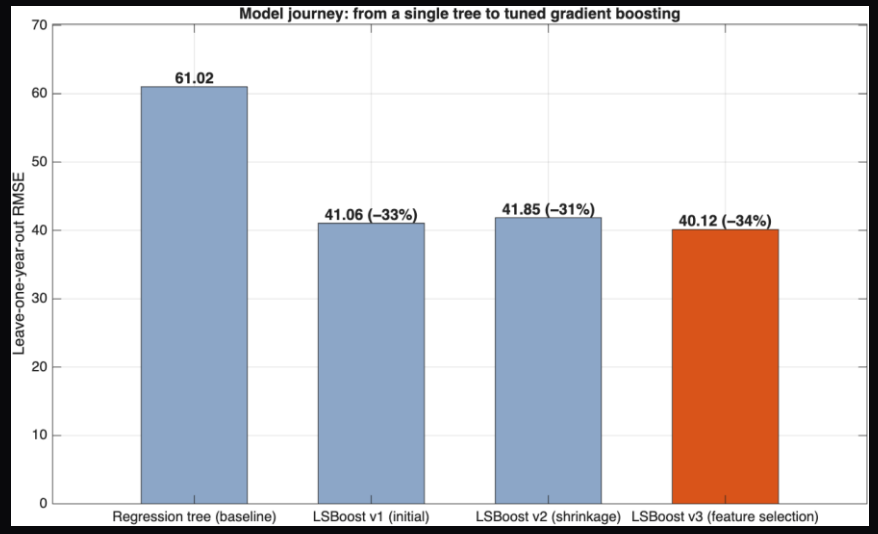
# Results

**LSBoost v3** is our final model:

- LOYO RMSE **40.1**, **34%** lower than a single tree
- MAE **21.0** kBtu/ft²/yr and $R^2$ **0.53** on years it never saw
- Best in **5 of 6** held-out years
- Reproducible: a clean rerun gives a byte-identical submission

Leave-one-year-out (official model)

| Model | All years | Year 6 | vs. tree |
|---|---|---|---|
| Tree | 61.02 | 59.37 | – |
| LSBoost v1 | 41.06 | 41.08 | −33% |
| LSBoost v2 | 41.85 | 42.57 | −31% |
| LSBoost v3 | 40.12 | 41.31 | −34% |



# Model Comparison

Controlled experiment: same data and folds, only the model changes.

- **M1**: LSBoost → single regression tree, all cleaned factors
- **M2** → single regression tree : LSBoost, 8 selected factors
- **M3**: LSBoost (300 Trees), 8 selected factors

Boosting on selected features wins on **every metric** under both validations, consistent with the official result.

Five-fold pooled out-of-fold

| Metric | M1 | M2 | M3 |
|---|---|---|---|
| MAE | 23.3941 | 23.3385 | 21.0390 |
| MSE | 2044.3723 | 2024.7375 | 1695.0047 |
| RMSE | 45.2147 | 44.9971 | 41.1704 |
| $R^2$ | 0.3976 | 0.4034 | 0.5005 |

Latest-year holdout (train years 1–5, predict 6)

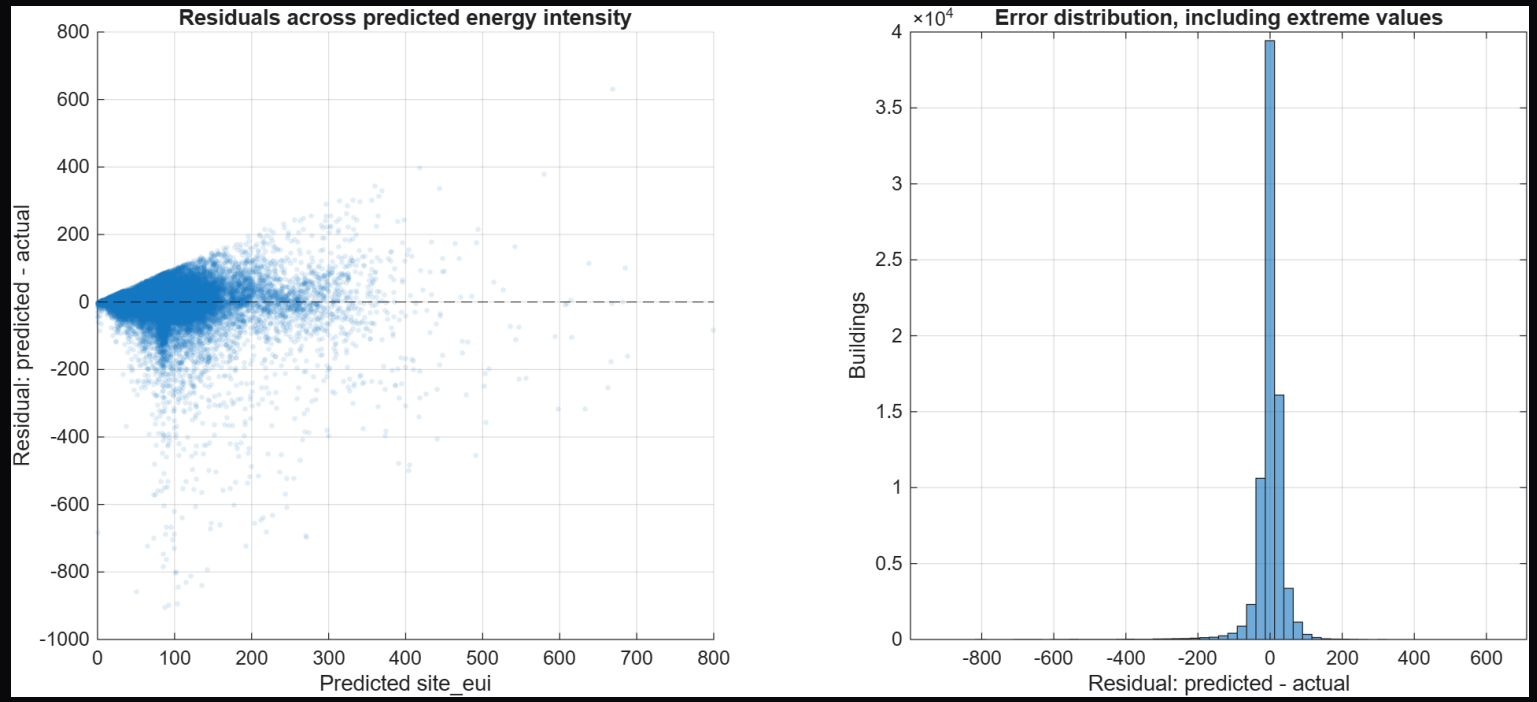
| Metric | M1 | M2 | M3 |
|---|---|---|---|
| MAE | 22.7927 | 22.7324 | 21.0440 |
| MSE | 2023.2171 | 2003.8279 | 1842.3056 |
| RMSE | 44.9802 | 44.7641 | 42.9221 |
| $R^2$ | 0.3777 | 0.3836 | 0.4333 |

# Results - Residual

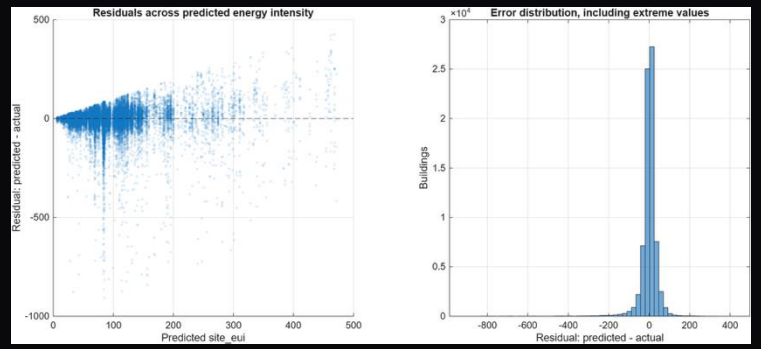
Residuals (predicted − actual) for all three models, five-fold out-of-fold:

- Centred on zero (M3 mean bias **0.03**)
- M3 has the tightest spread
- Long negative tail: a few buildings use **far more** than predicted

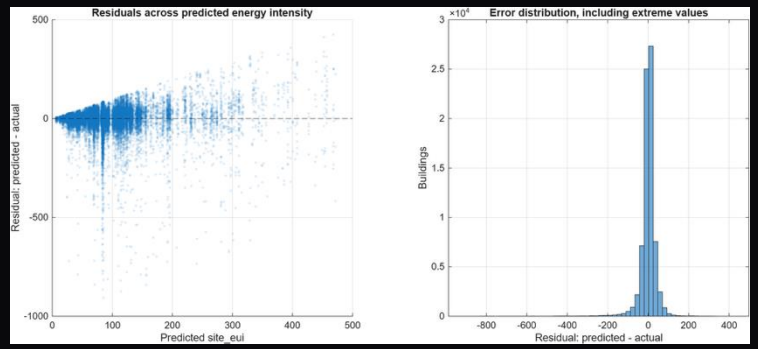
Those outliers are exactly what **Retrofit Radar** flags.




M3: LSBoost, selected factors (best)




M1: tree, all factors




M2: tree, selected factors

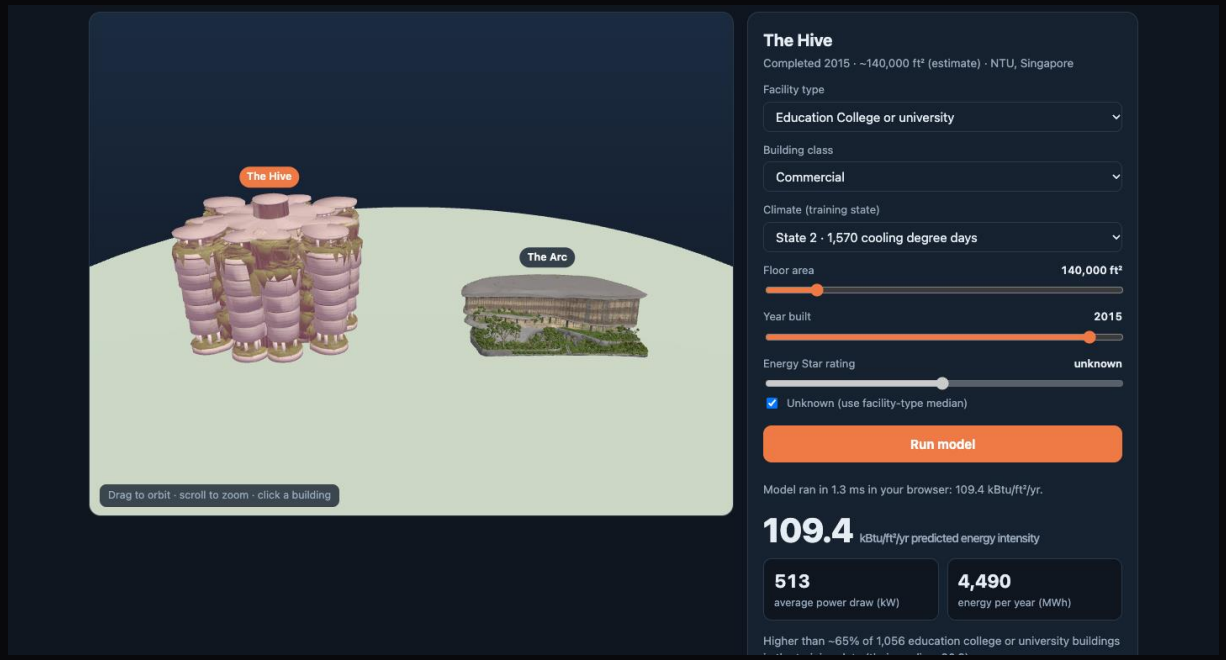
# Live Demo

**Retrofit Radar** compares each building's actual energy use with what the model expects for similar buildings:

- **1,316** buildings use 50%+ more than expected
- The top 50 account for **1,573 GWh/yr** of excess energy

**3D simulator**: click NTU's The Hive or The Arc and our trained model runs **in the browser**, matching MATLAB exactly.

Open **index.html** in the repository for the dashboard, simulator and report.




https://roh00t.github.io/MatrixNotFound/

# Conclusion

Our best model is **LSBoost v3 (feature selection)**: LOYO RMSE **40.1**, MAE **21.0**, $R^2$ **0.53**, with 34% lower error than a single tree.

In real-world use, **Retrofit Radar** turns predictions into a ranked audit list: the buildings using far more energy than their peers.

Repository link:

[Roh00t/MatrixNotFound: LearnToHack]

